

THEME • *CIRCULAR CARBON ECOSYSTEM*

Carbon-Aware Supply Chain Dashboard

Auditable Scope 3 accounting across multi-tier suppliers, plus a measure of what a company cannot see.

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Built for sustainability teams, supply chain managers, ESG auditors and suppliers

The emissions you cannot see

Visibility collapses one step past the suppliers a company pays directly.

70%+

of a typical footprint sits in the value chain, not owned operations.

2027

CSRD and ISSB deadlines make Scope 3 disclosure mandatory.

Most reduction budgets are spent above this line.

Tier 1
direct suppliers



Procurement visibility usually stops here

Tier 2
sub-suppliers



52.6% of the footprint sits here

Tier 3
raw processors



Tier 4
extraction



Worked through on aluminium

Aluminium is where the arithmetic gets interesting, because virgin and recycled material differ by roughly an order of magnitude.

Activity-based engine

Energy, materials and freight converted using published DEFRA, EPA and IEA factors, mapped to GHG Protocol categories 1, 3 and 4.

Multi-tier propagation

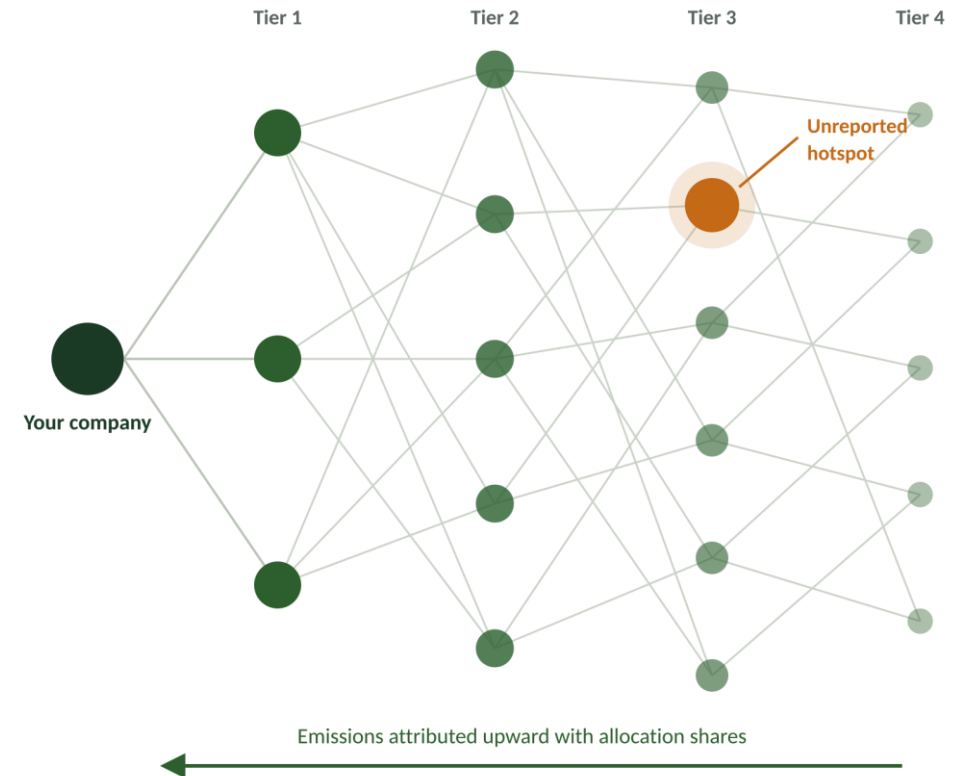
Deep-tier emissions attributed upward with explicit allocation shares.

Full audit trail

Click any figure to see how it was derived.

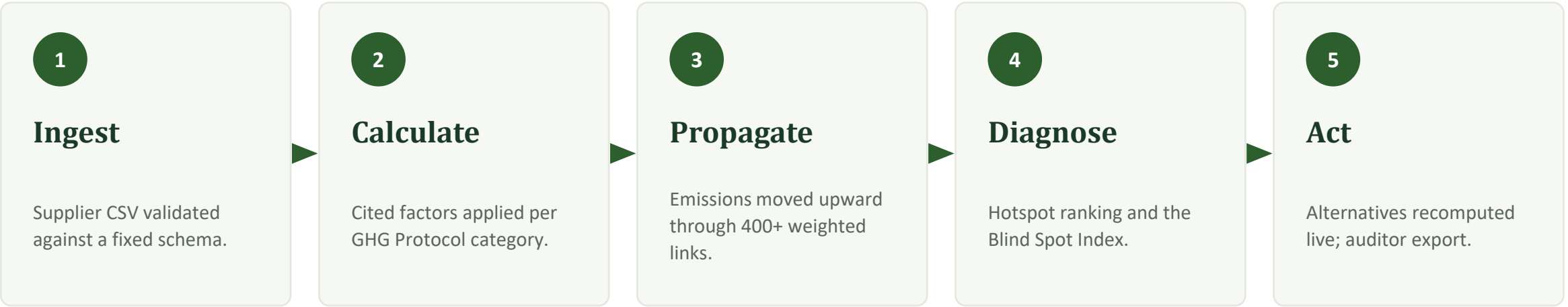
One smelter, 2,850 tonnes of aluminium

Virgin 35,625 tCO₂e → recycled 4,418 tCO₂e. Switching material and shifting air freight to sea removes 41,896 tCO₂e, or 47% of the whole chain.

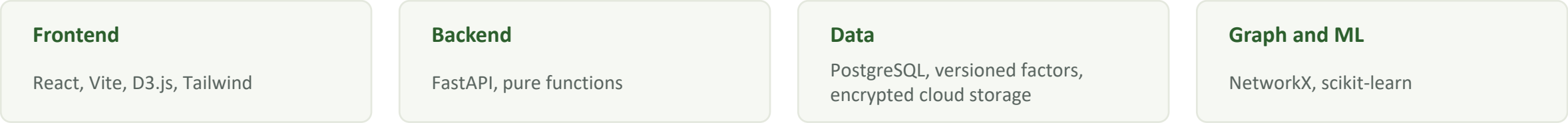


How it works

The calculation is deterministic. Machine learning only fills gaps in supplier data.



Stack



The Blind Spot Index

We could not find an existing tool that reports this, so we built the metric ourselves. It is the share of procurement spend running through suppliers that never reported.

41.2%

of spend unmeasured

- Line coverage reads 93.3%
- Yet 41.2% of spend is unverified
- Band: Weak, do not publish

From the prototype

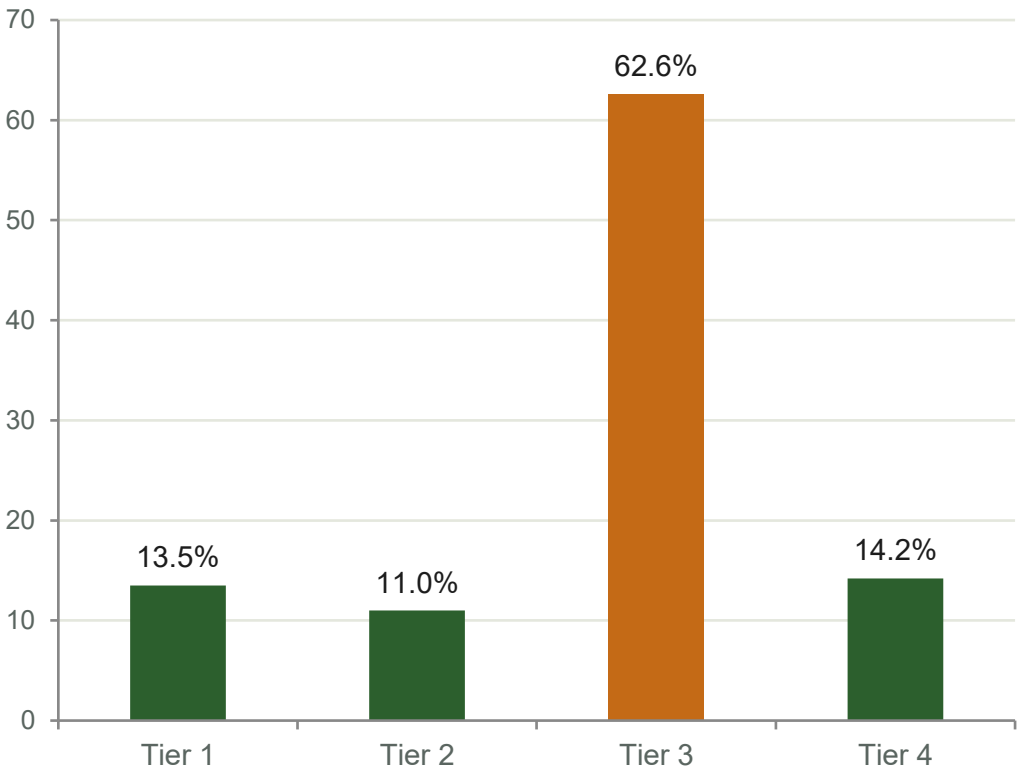
88,639

tCO2e across 200 suppliers and 413 supply links.

52.6%

of that comes from one supplier that reports nothing.

Blind Spot Index by supplier tier (%)



What could go wrong

Cost

Nothing in the stack carries a licence fee, and the published factor sets are free to use.

Adoption

CSV upload matches how suppliers already send data.

Value

Auditors can trace any figure back to its source, and managers can see which nodes are worth spending money on.

Principal risks and mitigations

Risk	Mitigation
Suppliers fail to report data	Gaps are quantified by the Blind Spot Index rather than hidden.
Factors differ by region and vintage	Factor tables are versioned and cited, so any report reproduces exactly.
Estimates mistaken for measurements	Imputed figures are visually distinguished everywhere.

The index falls as suppliers start reporting, which gives a company a reason to open the dashboard every quarter rather than once a year.

Where this goes next

Near term

Supplier self-service portal, driving the Blind Spot Index down each quarter.

Mid term

Marginal abatement cost curve ranking interventions by tonnes saved per unit of spend.

Longer term

ERP connectors replacing CSV upload, plus assurance of the audit export.

References

- GHG Protocol Corporate Value Chain (Scope 3) Standard (category definitions and boundary rules)
- UK DEFRA conversion factors (material intensities, freight per tonne-kilometre)
- US EPA eGRID, IEA and Australian National Greenhouse Accounts (grid electricity intensity)
- International Aluminium Institute (primary against recycled aluminium)
- CSRD and ISSB disclosure requirements (regulatory context)